

Program overview

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Year 2024/2025
Organization Aerospace Engineering
Education Pre-Master Aerospace Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
AE bridging programme 2024			
AE1205	Programming & Scientific Computing in Python for AE1	2	
AE2130-I S	Aerodynamics I	3	
AE2135-I S	Structural Analysis and Design	5	
AE2220-II S	Computational Modelling	3	
AE2230-I S	Flight and Orbital Mechanics	4	
AE2235	Aerospace Signals, Systems & Control	7	
AE3212-I S	Aerospace Flight Dynamics & Simulation	5	
IFEEMCS010400	Linear Algebra	5	
WI1402LR	Calculus II	5	
WI1421LR	Calculus I	6	
WI1909TH	Differential Equations	3	
WI2032TH	Numerical Analysis	3	

Year	2024/2025
Organization	Aerospace Engineering
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AE bridging programme 2024

AE1205	Programming & Scientific Computing in Python for AE1	2
Responsible Instructor	Dr.ir. J. Ellerbroek	
Instructor	Dr. G.C.H.E. de Croon	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Two 4 hour sessions per week (divided over 4 parts of the course , see below). Every session consists of instruction, examples, assignments and wrap-up/solutions. Also challenges and competitions are part of the course. Part 1 General Purpose Programming in Python Part 2 Scientific Computation & Plotting (NumPy, Matplotlib) Part 3 Numerical simulation, Visualisation & Animation (PyGame incl. game programming) Part 4 Contest & Diagnostic tests & Test Exam	
Study Goals	- translate a problem into an algorithm - basic programming skills - basics of Python+NumPy - numerical analysis - modeling - plotting data using matplotlib - simulation, visualisation & animation	
Education Method	Studio Classroom sessions, with: - Lecture (1 hour) - Exercises - (optional) assignments for 0.5 bonus point - (optional) competition for 0.5 bonus point - quizzes	
Computer Use	Bring your laptop with Python 3 installed	
Reader	Programming & Scientific Computing in Python for Aerospace Engineers	
Prerequisites	Install Python 3 (see Reader for instructions!)	
Assessment	Written/PC exam which contains multiple choice questions, assignments/open questions and fill in questions. Additionally one bonus point can be gained by meeting a certain quality in a final assignment, writing a simulation/game, and completing the session assignments. The bonus point is valid for the next four exams, so the exam and the resit of this year and the next.	
Exam Hours	0/0/0/3	
Permitted Materials during Tests	Non-programmable calculator or calculator in exam mode	
Set-up	Deliverables: week 1: Assignment week 2: Assignment week 3: Assignment week 4: Assignment week 5: Assignment week 6: project week 7: Assignment, test exam	

AE2130-I S	Aerodynamics I	3
Contact Hours / Week x/x/x/x	COURSE CODE ONLY VALID FOR BRIDGING PROGRAMME	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	For course description: see AE2130-I	
Study Goals	see AE2130-I	
Education Method	see AE2130-I	
Assessment	see AE2130-I	

AE2135-I S	Structural Analysis and Design	5
Contact Hours / Week x/x/x/x	THIS COURSE CODE IS ONLY VALID FOR BRIDGING PROGRAMME	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	see AE2135-I	
Study Goals	see AE2135-I	
Education Method	see AE2135-I	
Assessment	see AE2135-I	

AE2220-II S	Computational Modelling	3
Contact Hours / Week x/x/x/x	THIS COURSE CODE IS ONLY VALID FOR BRIDGING PROGRAMME	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	see AE2220-II	
Study Goals	see AE2220-II	
Education Method	see AE2220-II	
Assessment	see AE2220-II	

AE2230-I S	Flight and Orbital Mechanics	4
Contact Hours / Week x/x/x/x	THIS COURSE CODE IS ONLY VALID FOR BRIDGING PROGRAMME	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	see AE2230-I	
Study Goals	see AE2230-I	
Education Method	see AE2230-I	
Assessment	see AE2230-I	

AE2235	Aerospace Signals, Systems & Control	7
Module Manager	Dr.ir. M.M. van Paassen	
Contact Hours / Week x/x/x/x	See AE2235-I and AE2235-II	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module contains two courses, AE2235-I, Aerospace Systems and Control Theory and AE2235-II, Instrumentation and Signals. For a detailed overview of the contents, see these individual courses.	
Study Goals	At the end of this module, the student will know how to or be able to: - do signal acquisition, processing and filtering - apply frequency domain techniques to signal conditioning and processing - know building blocks and electronic components - apply signal modulation, produce a signal link budget - formulate linear system models, combine models, create control structures - determine dynamic properties of linear systems, identify desirable properties - evaluate stability properties of closed and open loop systems - tune control systems with classical tuning methods In summary, the student will have the tools and knowledge to work with linear dynamical systems and their input and output signals. In AE2235-I, the focus will be on systems modelling, in AE2235-II the focus will be on signal analysis. For a more detailed description of the study goals, consult the descriptions for AE2235-I and AE2235-II	
Education Method	For AE2235-I: - 14 oral lectures, 2 hours each - 7 e-learning modules, to be completed with Python with python-control installed, or with Matlab For AE2235-II: - 12 oral lectures, 2 hours each - a studio classroom session, which will include hands-on experimentation with hardware sensors and data acquisition equipment. At the session the data is analysed with Python or Matlab.	
Assessment	AE2235-I: Computer based exam, 2 hours. Sign up for the exam is through Osiris. During the exam, students have access to Matlab with the control toolbox and to Python and the python-control module. See BrightSpace sites for ae2235-I for information on the available editors in the exam environment, if you used other editors, plan on practicing a bit with editors in the exam environment. AE2235-II: Written exam, combination of open and multiple choice questions.	

AE3212-I S		Aerospace Flight Dynamics & Simulation	5
Contact Hours / Week x/x/x/x	THIS COURSE CODE IS ONLY VALID FOR BRIDGING PROGRAMME		
Education Period	3		
Start Education	3		
Exam Period	3 5		
Course Language	English		
Course Contents	see AE3212-I		
Study Goals	see AE3212-I		
Education Method	see AE3212-I		
Assessment	see AE3212-I		

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in $\mathbb{R}^n$ - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in $\mathbb{R}^n$ (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace.	
Assessment	Disclaimer: information may change depending on the developments around the coronavirus. Final score: F IF the midterm score M is higher than the score E for the exam AND $E \geq 5.0$ THEN $F = 0.25 M + 0.75 E$ ELSE $F = E$ All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

WI1402LR	Calculus II	5
Responsible Instructor	Dr. M.B. van Hoven	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Summary	Multivariable Calculus	
Course Contents	The goal of the course is to lay a strong foundation for the numerous and various applications of calculus, both directly within the field of aerospace engineering as in advanced engineering mathematics. The course focusses on both differentiation and integration techniques involving multivariable functions and vector fields. The main subjects of the course are: Techniques for partial and directional derivatives (tangent plane, linearisation, gradient vector, extreme values,) Integration techniques for functions of two or more variables. Double integrals using polar coordinates. Triple integrals using cylindrical and/or spherical coordinates. Line and surface integrals of functions and vector fields. Theorems of vector calculus by Green, Stokes and Gauss.	
Study Goals	At the end of this course, the student masters the course subjects at a variety of levels: Knowledge: reproduce and name the learned definitions, theorems and formulas. Comprehension: explain, interpret and derive the learned theorems and formulas. Application: apply the learned definitions, theorems and formulas in concrete mathematical problems. Analysis: decompose mathematical problems in relevant smaller (easier) subproblems. Synthesis: Composing strategies for solving a mathematical problem by combining several learned techniques. Evaluation: use learned mathematical tools to critically asses (and judge on) a given idea or proposition	
Education Method	A threefold workflow around each lecture: Preparation: Students study a pre-lecture video and/or do pre-lecture exercises to prepare for class. Participation: Students come to the lecture. An active attitude is required: taking part in practice exercises and poll questions posed by the lecturer. Asking questions. Practice and study: After the lecture the students practice exercises from the book and in GraspLe. Concepts discussed in class should be studied in detail (use the book).	
Literature and Study Materials	Adams, R.A. , "Calculus, a complete course", 10th ed., Addison Wesley (2021), ISBN 9780135732588	
Assessment	Written exam (two hours) at the end of the third quarter. The assessment contains short answer questions (only answer assessed) and open questions (calculation/motivation assessed) being a representative selection of the entire course content.	
Remarks	It is not allowed to use electronic devices such as calculators, mobile phones and laptops/notebooks etc. during the exam. Written notes and formulae sheets are also prohibited.	
Co-Instructor	Dr. W.M. Schouten-Straatman	

WI1421LR	Calculus I	6
Responsible Instructor	Dr. M.B. van Hoven	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Summary	Single Variable Calculus	
Course Contents	The goal of the course is to lay a strong foundation for the numerous and various applications of calculus, both directly within the field of aerospace engineering as in advanced engineering mathematics. The possibility of using computer algebra systems allows for a shift in emphasis from technical skills towards understanding the underlying concepts.	
Study Goals	Course subjects: - Vector arithmetic (addition, scalar multiplication, dot product, cross product). - Differentiation, linearisation, differentials. - Integration techniques for functions of one variable (e.g. substitution rule, integration by parts). - First-order differential equations by analytical methods. - Complex numbers, both in Cartesian and polar form (addition, multiplication, division, powers, etc.). - Second-order linear differential equations with constant coefficients. - Series and sequences: Convergence tests and techniques. - Taylor and Maclaurin series for the approximation of functions. - Introduction to multivariable functions: continuity and differentiability. At the end of this course, the student masters these subjects at a variety of levels: Knowledge: reproduce and name the learned definitions, theorems and formulas. Comprehension: explain, interpret and derive the learned theorems and formulas. Application: apply the learned definitions, theorems and formulas in concrete mathematical problems. Analysis: decompose mathematical problems in relevant smaller (easier) subproblems. Synthesis: Composing strategies for solving a mathematical problem by combining several learned techniques. Evaluation: use learned mathematical tools to critically assess (and judge on) a given idea or proposition	
Education Method	Lectures + seminars Typical routine: - Preparation: Students do homework, study a pre-lecture video (or do pre-lecture exercises) - Participation: Students come to the lecture which is a block of two hours. In the lecture a new topic will be introduced by the teacher. During the lecture there will be time to do exercises. Often a practical application of the theory is discussed. - Post-lecture work: Students do homework assignments. These contain book exercises as well as exercises in a digital environment.	
Literature and Study Materials	Adams, R.A. , "Calculus, a complete course", 10th ed., Addison Wesley (2021), ISBN 9780135732588	
Assessment	Preliminary examinations (two hours) at the end of the first and second quarters. Both assessments have a two-hour duration and consist of a number of short answer questions (only answer is assessed) and a number of open questions (calculation/motivation is assessed). *In the event of on-campus restrictions (e.g. due to COVID), assessments will be 90 min online (Grasple) proctored exams.	
Special Information		
Remarks	At the exam, it is not allowed to use electronic devices (such as calculators, mobile phones and laptops/notebooks etc.), formulae sheets.	
Co-Instructor	Dr. N.D. Verhulst	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

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